

Tasks

Objective Questions:

1. Data Collection and Cleaning: Import the necessary libraries, load the dataset, handle missing values, check data types, and ensure data is in the correct format for analysis.
2. Calculation of Recency, Frequency, and Monetary (RFM) values: Calculate the Recency, Frequency, and Monetary values for each customer.
3. Assign Scores to the RFM Values: Assign scores to the RFM values based on quantiles.
4. Create RFM Segments: Combine the RFM scores to create customer segments.
5. Customer Distribution Across Segments: Calculate and visualize the distribution of customers across different RFM segments.
6. Revenue Contribution by Segment: Determine and visualize the total revenue contribution of each RFM segment.
7. Recency Distribution: Visualize the distribution of recency scores among the customers.
8. Frequency Distribution: Visualize the distribution of frequency scores among the customers.
9. Monetary Value Distribution: Visualize the distribution of monetary values among the customers.
10. Top Segments by Customer Count: Find the RFM segments with the highest number of customers.

Subjective Questions:

1. How would you identify segments with low RFM scores indicating potential churn and what targeted interventions would you implement to retain these customers?

2. What marketing strategies would you design for each RFM segment? Focus on exclusivity and premium offers for high-value segments and use discounts and promotions for low-value segments.
3. Based on product preferences in each segment, how would you inform product development to enhance features of top-performing products and address gaps in the product line?
4. How would you leverage geographic data to create targeted marketing campaigns for regions with high-value customers and regions with low engagement?
5. How would you develop seasonal marketing campaigns based on the seasonal trends observed in different RFM segments to capitalize on peak purchasing times?
6. How would you identify cross-selling opportunities based on purchase data and create bundled offers for customers in different RFM segments?
7. Without direct feedback data, how would you use indirect feedback such as repeat purchases and product returns to refine RFM segmentation and improve customer satisfaction?
8. How would you analyze the revenue variation by day of the week for different RFM segments to identify patterns and optimize marketing efforts?